

QubitSky

Classical Baselines for Acoustic Drone Detection

Models A–D, Evaluation Protocol, and CPU/CUDA Execution

QubitSky Project

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Abstract

This report describes four classical baselines for distinguishing drone audio from background and acoustically similar confusers. Models A, B, and C use a shared compact feature representation with an RBF support vector machine, a small multilayer perceptron, and a larger multilayer perceptron, respectively. Model D uses a convolutional neural network operating on log-mel spectrograms. The protocol separates source recordings before windowing, fits preprocessing on training data, selects configurations on validation data, and evaluates noise robustness using controlled signal-to-noise ratios. Automatic CUDA detection allows Model D to use a compatible visible GPU or fall back to CPU. This is a methods document: it reports no new trained-model accuracy, runtime benchmark, or evidence of quantum advantage.

1 Purpose and implementation scope

The binary label is $y = 1$ for drone and $y = 0$ for non-drone. The objective is to establish reproducible classical reference points across model capacity and audio noise conditions. The models serve complementary roles:

Model	Input and classifier	Role in the comparison
A	Compact MFCC/PCA features; RBF SVM	Nonlinear kernel baseline
B	Same compact features; one small hidden layer	Limited-capacity neural baseline
C	Same compact features; two larger hidden layers	Higher-capacity tabular baseline
D	Log-mel spectrogram; three convolution layers	Reference using richer temporal information

Neither C nor D is guaranteed to achieve the best accuracy. Likewise, B is not claimed to match a quantum model’s parameter budget without an explicit count and agreed definition of capacity.

Snapshot boundary. The baseline described here follows the reviewed `qsky_classical/cli.py`, `data.py`, `models.py`, and `cnn.py` snapshot captured on October 7, 2026 at 20:23 UTC. File fingerprints accompany this document in `classical_approach_sources.json`. Protocol-import, recording-aggregation, and subset utilities were being developed concurrently; their presence is not treated as proof of an integrated, validated experiment. Section 9 distinguishes the comparison requirements from the baseline CLI behavior documented here.

2 Datasets and source attribution

The intended data sources follow `QubitSky_DATASETS.md`, `DATASET_SETUP.md`, and the repository’s downloader. Table 1 states their project roles; it does not assert that every dataset has been downloaded or used in a completed run. Actual membership, class counts, and exclusions must come from the run’s manifests and inventories.

Dataset	Published resource	Intended QubitSky role
DDL [2]	Field and synthesized drone audio; the downloader selects the real-recording archive	Main drone source; verify available labels before constructing the binary task
ESC-50 [3]	2,000 five-second clips across 50 environmental classes	Selected non-drone confusers and a separately reserved noise pool
UrbanSound8K [4]	8,732 urban-sound excerpts of at most four seconds across 10 classes	Selected urban/mechanical confusers and additive noise
NASA Small UAS [5]	Small unmanned-aircraft flyover acoustic recordings	External evaluation only; never training, validation, or preprocessing fitting

Table 1: Primary dataset sources and intended roles. Published corpus sizes are not the effective training-set sizes after filtering and grouping.

DDL: field recordings. Safavi et al.’s release provides separate real and synthesized archives [2]. The project selects `MLSP_2022_Real_Data.zip`. The publisher documents `MINI` and `PRO4` as drone classes and `XXXX` as no drone, together with flight/session and sequence identifiers. Consecutive fragments must be reconstructed in sequence within their source session before windowing; they must not be randomly split as independent recordings. The repository’s setup notes report no `XXXX` examples in the inspected real archive. Therefore, the naming convention alone does not establish that the selected archive supplies negative examples.

ESC-50 and UrbanSound8K: confusers and noise. The ESC-50 selection contains helicopter, airplane, wind, rain, thunderstorm, chirping birds, crow, crickets, insects, siren, footsteps, and fireworks. These 12 classes correspond to 480 published clips before any additional filtering [3]. UrbanSound8K contributes the selected classes engine idling, siren, car horn, gun shot, street music, drilling, and jackhammer [4]. These are project-specific binary selections, not the original 50-class or 10-class benchmark tasks.

Preserve ESC-50’s `src_file`, UrbanSound8K’s `fsID`, and the published fold metadata [3, 4]. Both corpora derive from Freesound: if the same source ID occurs in both, keep its excerpts in one split. Assign distinct source recordings to labeled confusers and additive-noise pools. Comparisons to published UrbanSound8K classification benchmarks require its prescribed ten-fold evaluation; this project-specific binary protocol is not that benchmark [4].

NASA: external transfer evaluation. The NASA portal provides the flyover archive and its data-description document [5]. The repository setup notes describe MATLAB recordings requiring conversion before waveform ingestion. Preserve acquisition metadata and keep all derived files external-only. Establish valid labels before reporting accuracy or F1. If labels are unavailable, report prediction summaries rather than external classification accuracy.

2.1 Additional sources in the teammate protocol

The previously reviewed teammate branch also references Fredrik Svanström’s drone-detection dataset [6] and the ITU ARIS Lab. Acoustic Drone Dataset [7]. These are distinct resources from the DDL real archive above. The ITU release contains overlapping one-second segments and source-interval mappings; its provided group assignments must be preserved rather than randomly splitting adjacent files. Reproducing the teammate’s experiment requires the exact dataset versions and source IDs in her manifests, not substitution based on a shared folder name such as `ddl`.

Dataset links and bibliographic details below were checked on October 7, 2026. Retain the source metadata and per-recording attribution with each downloaded release.

3 Data organization and audio preparation

3.1 Recording manifests and leakage prevention

A signal CSV provides `path`, `label`, and optionally `group_id` and `split`. Related excerpts from one recording, flight, or session must share a group identifier. Paths are resolved relative to the manifest, unless absolute. A separate noise manifest supplies additive-background recordings.

The default automatic split is approximately 60% training, 20% validation, and 20% test *by source groups*; it is stratified by group label. Unequal group sizes can produce different proportions of files or windows. Explicit splits must cover every row, keep groups disjoint, and contain both labels in each signal split. Mixed-label groups require explicit assignments. The data reader also recognizes `validation` and `valid` as `val`.

Splitting occurs before audio is expanded into windows. The baseline runner rejects shared file paths or source-group identifiers between the signal and noise manifests. Training augmentation uses only training noise, and test corruption uses only test noise. External NASA data, if used, must remain outside training and validation; the generic CSV interface relies on the manifest preparer to enforce that dataset role.

3.2 Waveform preparation

Each recording is converted to mono and resampled to $f_s = 16,000$ Hz. The reviewed baseline CLI uses non-overlapping one-second windows, each containing $N = 16,000$ samples. The final incomplete window is zero-padded rather than discarded. Each window x is peak-normalized:

$$\tilde{x}[n] = \frac{x[n]}{\max(\max_m |x[m]|, 10^{-12})}. \quad (1)$$

This removes absolute amplitude as a direct cue. A short padded tail still contributes a full window, so its influence should be considered during recording-level aggregation. The current data helpers accept other durations; the reviewed CLI still invokes their one-second default.

4 Controlled noise experiments

For target window x and background window n , define

$$\text{RMS}(x) = \sqrt{\frac{1}{N} \sum_{i=1}^N x[i]^2}. \quad (2)$$

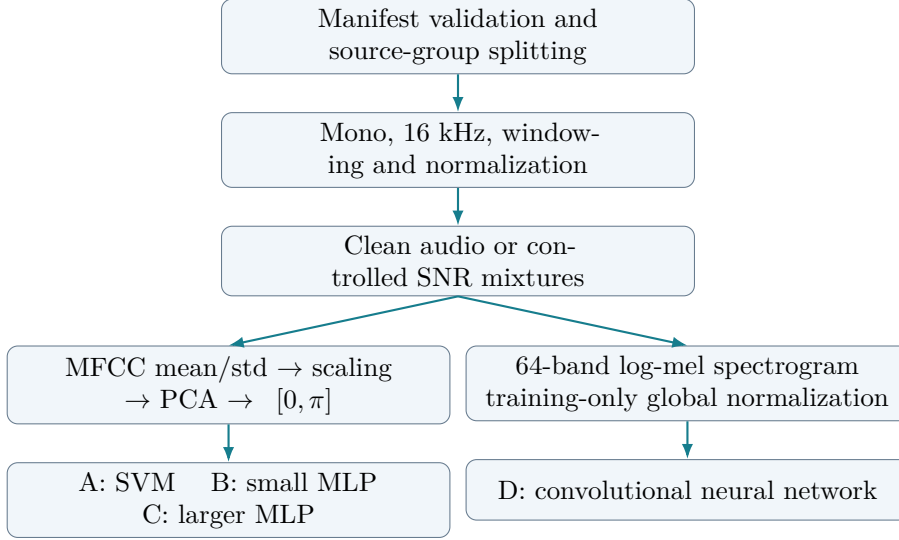


Figure 1: Shared audio conditions feed two feature branches. Fitted transforms use training data only; validation and test data never refit them.

To achieve a requested SNR s in decibels, the mixture is

$$x_s = x + \alpha n, \quad \alpha = \frac{\text{RMS}(x)}{\text{RMS}(n)} 10^{-s/20}. \quad (3)$$

Thus $s = 10 \log_{10}(\|x\|_2^2 / \|\alpha n\|_2^2)$ for nonzero signal and noise. The mixture remains floating-point audio and is not clipped or renormalized after mixing. At 0 dB, the target and added noise have equal power; at −10 dB, the added noise has ten times the target power.

The baseline evaluates clean, 20, 10, 0, and −10 dB conditions by default. The `--snrs` argument also supports the teammate protocol’s clean, 20, 10, 5, and 0 dB conditions. For a fixed seed, each target receives the same donor across SNR levels and models; only its gain changes. Optional augmentation adds noisy training copies alongside the original clean windows. Clean validation remains the baseline selection condition.

Silence policy. SNR is undefined if either waveform has zero power. The reviewed runner filters silent noise windows but raises an error for a silent target in a noisy condition. Clean-only evaluation can retain silence. A helper for recording silent-window exclusions exists in the evolving data module, but the reviewed CLI does not yet invoke it. Before a matched benchmark, agree on and log an exclusion policy that preserves comparable evaluated recordings across conditions; do not assign a fabricated SNR to silence.

5 Feature representations

5.1 Models A–C: compact MFCC statistics

For each window, a power mel spectrogram uses 64 mel bands, a 512-sample FFT, and a 160-sample hop. The power is converted to decibels with reference value 1.0, and 13 MFCCs are computed from that log-mel representation using `librosa` [8]. For coefficient j over T frames,

$$\mu_j = \frac{1}{T} \sum_{t=1}^T c_{j,t}, \quad \sigma_j = \sqrt{\frac{1}{T} \sum_{t=1}^T (c_{j,t} - \mu_j)^2}. \quad (4)$$

Concatenating the 13 means and 13 population standard deviations produces $\mathbf{f} \in \mathbb{R}^{26}$. The preprocessing sequence is

$$\mathbf{f} \xrightarrow{\text{StandardScaler}} \mathbf{z} \xrightarrow{\text{PCA}} \mathbf{u} \in \mathbb{R}^k \xrightarrow{\text{MinMaxScaler}} \boldsymbol{\theta} \in [0, \pi]^k, \quad k \in \{2, 4, 6\}. \quad (5)$$

All three transforms are fitted on training windows only, including noisy training copies when augmentation is enabled. PCA uses the full SVD solver. Held-out values outside training extrema are clipped to $[0, \pi]$. The exact transformed arrays are exported for possible quantum reuse. Angle scaling is an interface choice; it does not establish that these values match another pipeline’s selected acoustic features.

5.2 Model D: log-mel spectrograms

D retains the 64-band log-mel matrix instead of reducing it with PCA. Under the baseline’s centered framing, a one-second window produces a 64×101 matrix. A single mean and standard deviation, computed over all training spectrogram entries, normalize training and held-out inputs:

$$\widehat{M} = \frac{M - \mu_{\text{train}}}{\max(\sigma_{\text{train}}, 10^{-6})}. \quad (6)$$

D retains time-frequency structure that A–C discard. It is therefore a richer-input reference, not a comparison constrained to the same k features.

6 Models and validation selection

6.1 A: radial-basis-function support vector machine

The SVM uses scikit-learn’s SVC with kernel

$$K(\boldsymbol{\theta}_i, \boldsymbol{\theta}_j) = \exp\left(-\gamma \|\boldsymbol{\theta}_i - \boldsymbol{\theta}_j\|_2^2\right). \quad (7)$$

It searches $C \in \{0.1, 1, 10\}$ and $\gamma \in \{\text{scale}, 0.1, 1\}$, giving nine candidates per PCA width. Class weights are balanced. The decision-function score, not a calibrated probability, is used for evaluation, with threshold zero [9]. Support-vector counts are recorded; they are not directly equivalent to neural network parameter counts.

6.2 B and C: multilayer perceptrons

B uses one ReLU hidden layer of width $h = 4$ by default, configurable through `--small-width`. C searches hidden widths (64, 32) and (128, 64). Both use a binary logistic output, L-BFGS optimization, and $\alpha \in \{10^{-4}, 10^{-2}\}$ for L2 regularization. The default maximum iteration count is 500. There is no random internal early-stopping split; selection uses the externally reserved validation recordings [10]. Unlike A and D, the baseline MLP fitting call does not apply class weights.

For layer widths $d_0 = k, d_1, \dots, d_L = 1$, the number of weights and biases is

$$P = \sum_{\ell=1}^L (d_{\ell-1} + 1) d_{\ell}. \quad (8)$$

For B, $P = h(k + 2) + 1$. Its default width therefore gives 17, 25, and 33 parameters at $k = 2, 4, 6$. The two C architectures contain $64k + 2177$ and $128k + 8449$ parameters, respectively. Predictions use the positive-class probability with threshold 0.5. Optimizer convergence warnings must be reviewed before treating a fitted candidate as a completed benchmark.

6.3 D: convolutional neural network

All convolutions use 3×3 kernels, stride one, and padding one. The architecture contains 25,409 trainable parameters, including biases:

Stage	Operation	Parameters
1	Conv $1 \rightarrow 16$, ReLU, 2×2 max pooling	160
2	Conv $16 \rightarrow 32$, ReLU, 2×2 max pooling	4,640
3	Conv $32 \rightarrow 64$, ReLU	18,496
4	Adaptive average pooling to 1×1 ; flatten	0
5	Linear $64 \rightarrow 32$, ReLU, dropout 0.2	2,080
6	Linear $32 \rightarrow 1$	33
Total	One output logit; sigmoid used for inference	25,409

The optimizer is Adam with learning rate 10^{-3} and weight decay 10^{-4} . Defaults are 30 epochs and batch size 32. The binary cross-entropy loss weights the positive class by $w_+ = N_0/N_1$, using training-window label counts:

$$\mathcal{L} = -\frac{1}{B} \sum_{i=1}^B [w_+ y_i \log \sigma(a_i) + (1 - y_i) \log(1 - \sigma(a_i))], \quad (9)$$

implemented with `BCEWithLogitsLoss` for numerical stability. The best validation checkpoint is retained over the requested epochs; this is checkpoint selection, not patience-based early stopping.

6.4 Common selection rule

The reviewed CLI selects candidates using clean validation *clip-level balanced accuracy*. Ties retain the first candidate or earliest CNN epoch. Validation data are not merged into training. A tabular helper also supports F1-based selection with balanced accuracy and recall as tie-breakers, but the reviewed CLI uses its default balanced-accuracy rule. Every requested PCA width is reported rather than selecting a width by test performance.

7 Evaluation and interpretation

For file r with m_r window scores, the baseline aggregates

$$\bar{s}_r = \frac{1}{m_r} \sum_{j=1}^{m_r} s_{r,j}, \quad \hat{y}_r = \mathbf{1}\{\bar{s}_r \geq \tau\}, \quad \tau = 0 \text{ for A, } 0.5 \text{ for B–D.} \quad (10)$$

Results are emitted at both window and clip levels. In this runner, *clip ID means a file path*: source-group IDs prevent leakage during splitting, but multiple files from one source group are not automatically merged into one scored recording. This distinction matters for a fair quantum comparison.

The reportable metrics include accuracy, balanced accuracy, drone precision, drone recall, F1, specificity, ROC-AUC, and confusion counts. In particular,

$$\text{BA} = \frac{1}{2} \left(\frac{TP}{TP + FN} + \frac{TN}{TN + FP} \right), \quad (11)$$

$$F_1 = \frac{2TP}{2TP + FP + FN}. \quad (12)$$

The implementation returns zero for undefined precision/recall/F1 divisions and reports ROC-AUC as unavailable when only one label is present. Accuracy alone can hide poor drone recall in an imbalanced test set.

Training time includes candidate fitting and validation. Prediction time measures window scoring, excluding waveform preparation, feature extraction, and clip aggregation. A device comparison must therefore report preprocessing and end-to-end wall time separately. No runtime estimates in this document should be interpreted as hardware measurements.

8 CPU and CUDA execution

Models A–C and audio feature extraction run on CPU. Model D supports `--device auto`, `--device cpu`, and `--device cuda`. In automatic mode the implementation queries the visible CUDA devices and attempts a small allocation and kernel operation on each, selecting the first usable device. If none works, it selects CPU; probe failures generate a warning. An explicit CUDA request instead fails early when no usable device is available.

Selection respects the devices exposed by the environment, including `CUDA_VISIBLE_DEVICES` under a cluster scheduler. L40S, A100, and newer compatible NVIDIA GPUs can be used with a suitable CUDA-enabled PyTorch build; there is no A100-specific architecture or minimum-model-name restriction. The fallback is CPU rather than automatic Apple MPS selection. A CPU-only PyTorch installation cannot use an NVIDIA GPU simply because one is installed.

The CNN, loss weight, and mini-batches move to the selected device. Training uses pinned host memory for CUDA batches; inference returns scores to CPU before conversion to NumPy. Saved CNN state dictionaries contain CPU tensors, allowing a GPU-trained model to be loaded on a laptop. The requested device, resolved device, and device name are recorded in the run configuration or model metadata. A mid-run out-of-memory failure is reported rather than silently restarting the experiment on CPU.

The code seeds PyTorch, requests deterministic algorithms, disables cuDNN benchmarking on CUDA, and sets `CUBLAS_WORKSPACE_CONFIG` to `:4096:8` if it was unset. CPU training retains the original single-thread setting. These choices help reproducibility; they do not guarantee identical results across hardware or software versions [11].

9 Requirements for a matched quantum comparison

The teammate experiment and this baseline cannot be compared solely by model name or by the number of input features. The teammate details below refer to the previously reviewed **dhanya** branch snapshot **f31b415**, rather than later frontend revisions. The following protocol must be frozen before a new matched benchmark:

1. **Exact examples and labels.** Reuse the intended 24 training, 80 validation, and 98 test recording IDs for reproduction of the initial comparison. Obtain the source tables and audio; do not reconstruct membership from reported counts. D requires the matching audio, not just tabular features.
2. **Audio and noise.** Agree on duration (including the teammate’s three-second segmentation), padding, normalization, donor recordings, and clean/20/10/5/0 dB conditions. Log silent-window exclusions and donor provenance.
3. **Representation.** Reproducing the existing quantum experiment requires its exact selected acoustic features, feature order, and fitted transformations. The baseline’s MFCC/PCA coordinates are a

different representation even when their dimensions agree. D should be presented as a richer-input reference.

4. **Aggregation.** The teammate quantum pipeline averages feature vectors per recording before classification. This differs from averaging predictions after classification:

$$f\left(\frac{1}{m}\sum_j \mathbf{x}_j\right) \neq \frac{1}{m}\sum_j f(\mathbf{x}_j) \quad \text{in general for nonlinear } f.$$

Match feature aggregation for A–C, including its order relative to any nonlinear or clipped preprocessing, and explicitly document D’s scoring rule.

5. **Label budget.** The previously reviewed selected-feature metadata used 301 training recordings for supervised ranking, although downstream models used smaller subsets. Reproducing that protocol and measuring strict 24-recording label scarcity are different questions. For strict scarcity, feature selection and preprocessing must be fitted inside each actual training subset; additional noise or development data must be disclosed.
6. **Selection and repeated runs.** Choose hyperparameters and checkpoints on validation only. Use identical subsets across models for each seed, preserve a fixed held-out test set, and report variability across seeds. Corrected classical preprocessing also requires rerunning the quantum models; old quantum scores do not establish a matched comparison to a new protocol.

The evolving workspace includes import, subset, feature-aggregation, and artifact utilities to support this work. Their results are not claimed here as validated experiments. Full training, quantum reruns, and frontend integration remain separate tasks.

10 Artifacts, execution, and reporting status

10.1 Baseline output contract

Each run uses a new or empty output directory. The reviewed runner writes:

- `config.json`, `splits.csv`, and, when needed, `noise_splits.csv`: arguments, recorded versions, audio/device settings, and exact file/group split assignments.
- `selection.json`: validation candidates, selected hyperparameters or CNN epoch, and model-size information.
- `results.csv` and `predictions.csv`: condition-level metrics and individual scores, labels, and predictions.
- `artifacts/model_A_k*.joblib`, corresponding B/C bundles, and `artifacts/model_D.pt`: fitted models and preprocessing information.
- `artifacts/preprocessor_k*.joblib` and `artifacts/features_k*.npz`: preprocessing and exact transformed feature arrays, labels, file IDs, window IDs, and training SNR labels.

For publication, additionally archive dataset provenance, source-group-level evaluation IDs, checksums, the exact environment, and exclusion logs. Retain compatible scikit-learn environments for serialized estimators: the previously reviewed teammate frontend artifacts recorded version 1.9.1, whereas this package snapshot specifies `scikit-learn>=1.4,<1.8`. CPU-portable tensor storage addresses device portability for D, not every software-version issue.

10.2 Commands for a later authorized run

The following commands are examples, not evidence that a run has occurred. They use the documented baseline interface and are not a reproduction of the teammate’s complete three-second, selected-feature protocol.

```
# A-C: baseline MFCC/PCA experiment
python -m qsky_classical --manifest data/clips.csv \
  --noise-manifest data/noise.csv --models A B C \
  --pca 2 4 6 --snrs clean 20 10 5 0 --output runs/tabular

# D: automatic CUDA selection with CPU fallback
python -m qsky_classical --manifest data/clips.csv \
  --noise-manifest data/noise.csv --models D \
  --snrs clean 20 10 5 0 --device auto --output runs/cnn

# Lightweight device/inference tests; no fitting or training
python -m pytest -q tests/test_devices.py
```

Evidence boundary. The preceding device-change verification recorded 21 passing non-training checks, one skipped real-CUDA check, and three deliberately excluded fitting/training tests. This describes that earlier verification run, not a fresh validation of concurrent protocol changes. Creating this report launches no model training. The synthetic demo dataset is a functional smoke test, not a measure of drone-detection performance. Real-data accuracy, noise curves, seed variability, CUDA training behavior, and hardware runtimes still require dedicated experiments.

References

- [1] QubitSky Project. *Classical implementation source snapshot*. `qsky_classical/cli.py`, `data.py`, `models.py`, and `cnn.py`; dataset-role notes in `QubitSky_DATASETS.md` and `DATASET_SETUP.md`. October 7, 2026. See `classical_approach_sources.json` for SHA-256 fingerprints.
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